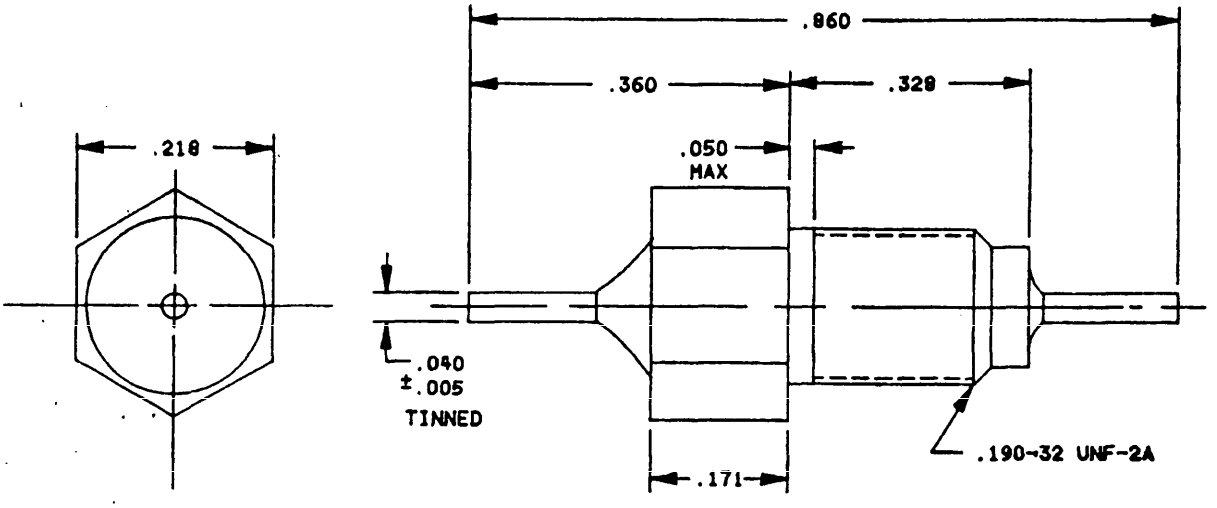
MILITARY SPECIFICATION SHEET

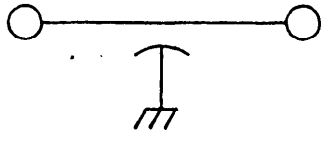
CAPACITORS, FIXED, FEED THROUGH, ELECTROMAGNETIC INTERFERENCE SUPPRESSION HERMETICALLY SEALED IN METAL CASES

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-83439.



INCHES	MM
.005	0.13
.020	0.51
.040	1.02
.050	1.27
.171	4.34
.190	4.83
.328	8.33
.360	9.14
.860	21.84



CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Mounting hardware shall not be supplied with the capacitors.
4. Unless otherwise specified, tolerance shall be ±.020 (0.51 mm).

FIGURE 1. Case dimensions and circuit configuration.

FSC 5910

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:07Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics.

Military part number M83439/08-	Capacitance	Capacitance tolerance	Rated voltage		Minimum insertion loss (dB)							
			dc	ac	50 kHz	100 kHz	500 kHz	1 MHz	10 MHz	100 MHz	1 GHz	10 GHz
	µF	%	volts	volts								
001	1.0	+80,-20	80	50	15	20	30	40	52	70	78	80

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Material:
Case: Steel, tin-plated.
Terminals: Solderable.
Dielectric: Ceramic.

Temperature range: -55°C to +125°C.

Rated voltage: See table I.

Capacitance: See table I.
Test frequency: 1,000 ±100 hertz.

Capacitance tolerance: See table I.

Terminal strength: Method 211 of MIL-STD-202, test condition A.
Applied force: 5 pounds -0, +3 ounces.

Insulation resistance: Method 302 of MIL-STD-202, test condition A, 10,000 megohms minimum.

Dissipation factor: Not more than 3.0 percent.

Insertion loss (abbreviated): See table I.

Voltage drop: 0.04 volt maximum.

DC resistance: 0.004 ohm maximum.

Rated current: 10.0 amperes maximum.

Immersion: In accordance with MIL-C-83439.
Capacitance change: Not more than ±20 percent.

Moisture resistance: In accordance with MIL-C-83439.

Capacitance change: Not more than ±20 percent from the initial measurement.

Capacitance change at reduced temperature: Not more than the capacitance tolerance specified, see table I.

Life at +85°C: Method 108 of MIL-STD-202, test condition D.
Operating conditions: 1.4 times dc rated voltage applied.
Measurements after exposure:
Insulation resistance
Capacitance change: Not more than the capacitance tolerance specified see table I.

Part number: See table I.

Custodian:
Air Force - 85

Review activities:
Air Force - 11, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 85

(Project 5910-F395)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:07Z
Check the source to verify that this is the current version before use.